

# Every Algebraic Number is a Critical Value of an Integer Polynomial

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## Abstract

We arrived at an independent proof of Petrov's theorem. For every irreducible  $f \in \mathbb{Z}[x]$  of positive degree there are  $g \in \mathbb{Z}[x]$  and an irreducible  $F \in \mathbb{Z}[x]$  of positive degree with  $F \mid g'$  and  $F^2 \mid f \circ g$ . Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of  $\mathbb{Z}[x]$  is exactly  $\overline{\mathbb{Q}}$ . The construction is explicit. We classify  $g \in \mathbb{Z}[x]$  with critical value  $\alpha$ , and find a lower bound for  $\deg g$ . The proof was assisted by AI, and has been formalized in Lean.

## 1 Introduction

A critical value of a nonconstant  $g \in \mathbb{Z}[x]$  is a number  $\alpha$  with  $g(\beta) = \alpha$  and  $g'(\beta) = 0$  for some  $\beta$ . Since  $\overline{\mathbb{Q}}$  is closed under polynomial evaluation, every critical value is algebraic. Schwarz, Vologodsky and Walcher posed the problem of establishing the reverse implication [SVW17]. This note solves the problem.

**Petrov's Theorem.** *Let  $f$  be an irreducible integral polynomial of positive degree. Then there exist  $F$  and  $g$  in  $\mathbb{Z}[x]$ , with  $F$  irreducible and of positive degree, such that*

$$F \mid g' \quad \text{and} \quad F^2 \mid f \circ g.$$

The construction in §2 was derived independently. Petrov's theorem [Pet21] was found later. The corollary is the statement in the title.

**Corollary 1.** *The set of critical values is exactly  $\overline{\mathbb{Q}}$ .*

**pf.** Write  $f = m_\alpha$ . If  $g(\beta) = \alpha$  and  $g'(\beta) = 0$  then  $\beta$  is a double root of  $f \circ g$ , since  $(f \circ g)' = (f' \circ g)g'$ , so  $m_\beta \mid g'$  and  $m_\beta^2 \mid f \circ g$ . Conversely take  $F$  and  $g$  from Petrov's theorem. A root  $\beta$  of  $F$  has  $g'(\beta) = 0$  and  $f(g(\beta)) = 0$ , so  $g(\beta)$  is a conjugate of  $\alpha$ , and conjugation gives  $\alpha$  itself. ■

## 2 The construction

Let  $f$  be irreducible of degree  $n \geq 1$  and  $\alpha$  a root. Write  $f(x) = \sum_{k=0}^n a_k x^{n-k}$ , so that  $a_n = f(0)$ . If  $f = x$ , then  $g = x^2$  and  $F = x$  work. Otherwise  $f(0) \neq 0$ , and since  $\gcd(f, f') = 1$  in  $\mathbb{Q}[x]$ , the elimination ideal  $(f, f') \cap \mathbb{Z}$  is nonzero. Let  $u, v \in \mathbb{Z}[x]$  and  $\rho \in \mathbb{Z}, \rho \neq 0$ , be such that  $u f + v f' = \rho$ , and set  $M := \rho a_n$ ,  $F(x) := f(Mx)/a_n$  and  $W(x) := (v(0)F(x) - v(Mx))/\rho$ . The polynomial claimed by Petrov's theorem is

$$g(x) := Mx + f(Mx)W(x). \tag{1}$$

**pf.** Write  $F(x) = \sum_{k=0}^n A_k x^{n-k}$ . From  $A_k = a_k M^{n-k}/a_n$  we get  $A_n = 1$  and  $A_k = a_k \rho^{n-k} a_n^{n-k-1}$  for  $k < n$ , so  $F \in \mathbb{Z}[x]$  with  $F(0) = 1$ , which gives  $W(0) = 0$ . And  $\rho$  divides  $M$  and every  $A_k$  with  $k < n$ , so it divides  $v(0)F(x) - v(Mx)$  and  $W \in \mathbb{Z}[x]$ . Therefore  $g$  is an integral polynomial.

Let  $\beta = \alpha/M$ , a root of  $F$ . Therefore  $g(\beta) = \alpha$ , and  $\rho W(\beta) = -v(\alpha)$ . Then

$$\begin{aligned} g'(\alpha/M) &= M + \rho a_n f'(\alpha) W(\alpha/M) + \cancel{f(\alpha)} W'(\alpha/M) \\ &= M - a_n f'(\alpha) v(\alpha) = 0, \end{aligned}$$

because  $u(\alpha) \cancel{f(\alpha)} + v(\alpha) f'(\alpha) = \rho$ . Hence  $\beta$  is a double root of  $f \circ g$ , and  $F$  is irreducible, so  $F \mid g'$  and  $F^2 \mid f \circ g$  by Gauss. ■

Any  $\rho$  will do, but  $M = \rho a_n$  multiplies into every coefficient of  $F$ ,  $W$  and  $g$ , so the reduced resultant gives the answer with the smallest coefficients.

### 3 Classification

We say a polynomial  $p$  is reducible if it is not primitive, or  $p = fg$  with  $\deg f, \deg g < \deg p$ , and irreducible otherwise. For  $\alpha \in \overline{\mathbb{Q}}$  we normalize the minimal polynomial  $m_\alpha$  with positive leading coefficient for uniqueness. Say that  $\alpha$  is a critical value for  $\beta \in \overline{\mathbb{Q}}$  if some  $g \in \mathbb{Z}[x]$  has  $g(\beta) = \alpha$  and  $g'(\beta) = 0$ , and call such a  $g$  a solution for  $\beta$ . Here we classify which  $g$  can appear for a given  $\alpha$ . Let  $P \in \mathbb{Z}[x]$  with  $\deg P > 0$ . By the fundamental theorem of algebra,  $P(x) - \alpha$  has a root  $\beta$ . Moreover this root is algebraic because it is a root of  $m_\alpha \circ P \in \mathbb{Z}[x]$ .

**Theorem 1.** *Let  $P \in \mathbb{Z}[x]$  be nonconstant, let  $\beta$  be a root of  $P - \alpha$ , and write  $m_\alpha \circ P = m_\beta h$  with  $h \in \mathbb{Z}[x]$ . Then  $\alpha$  is a critical value for  $\beta$  if and only if*

$$m'_\alpha(\alpha) \mid h(\beta) \text{ in } \mathbb{Z}[\beta].$$

*In that case, for  $Q \in \mathbb{Z}[x]$  with  $Q(\beta) = -h(\beta)/m'_\alpha(\alpha)$ ,  $g_0 = P + m_\beta Q$  is a solution for  $\beta$ , and the solutions  $g$  for  $\beta$  are exactly  $g \in g_0 + m_\beta^2 \mathbb{Z}[x]$ .*

*pf.* By Gauss's lemma  $m_\beta$  divides  $m_\alpha \circ P$  in  $\mathbb{Z}[x]$ , since  $m_\beta$  is irreducible. Differentiating  $m_\alpha \circ P = m_\beta h$  at  $\beta$  gives  $m'_\alpha(\alpha)P'(\beta) = m'_\beta(\beta)h(\beta)$  with  $m'_\alpha(\alpha), m'_\beta(\beta) \neq 0$ , so the two divisibilities are equivalent and  $-h(\beta)/m'_\alpha(\alpha) = -P'(\beta)/m'_\beta(\beta)$ . Now  $g_0(\beta) = P(\beta) = \alpha$  and  $g'_0(\beta) = P'(\beta) + m'_\beta(\beta)Q(\beta) = 0$ . If  $g$  is a solution for  $\beta$  then  $g - P$  vanishes at  $\beta$ , so  $g = P + m_\beta q$  with  $q \in \mathbb{Z}[x]$ , and  $0 = g'(\beta) = P'(\beta) + m'_\beta(\beta)q(\beta)$  gives the divisibility and  $q(\beta) = Q(\beta)$ , so  $m_\beta \mid q - Q$  and  $g \in g_0 + m_\beta^2 \mathbb{Z}[x]$ ; conversely every  $g_0 + m_\beta^2 r$  has value  $\alpha$  and derivative 0 at  $\beta$ . ■

Since the solutions for  $\alpha$  are cosets  $g \in g_0 + m_\beta^2 \mathbb{Z}[x]$ ,  $\deg g$  is unbounded above. So the natural question is the smallest  $\deg g$  for  $\alpha$ ; we do not answer that question, but we provide the following lower bound.

**Proposition 1.** *If  $g$  is a nonconstant solution for  $\beta$ , then  $\deg g > \deg \alpha$ .*

*pf.* Since  $\alpha = g(\beta)$ ,  $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\beta)$ , so  $\deg \beta \geq \deg \alpha$ . And  $g' \neq 0$  vanishes at  $\beta$ , so  $m_\beta \mid g'$  and  $\deg g - 1 \geq \deg \beta$ . ■

Each line below is a closed identity, checkable by expansion.

*Example 1* ( $f = qx - p$ ,  $p \neq 0$ ,  $\gcd(p, q) = 1$ ). Here  $f' = q$  and  $p = qx - f$ , so  $(f, f') \cap \mathbb{Z} = \mathbb{Z}$  and  $\rho = 1$ . Choose  $a, b \in \mathbb{Z}$  with  $ap + bq = 1$  and take  $u = -a$ ,  $v = ax + b$ . Then  $f(0) = -p$ ,  $M = -p$ ,  $F = qx + 1$ ,  $W = x$ , and

$$g = -pqx^2 - 2px, \quad g' = -2p(qx + 1), \quad f \circ g = -p(qx + 1)^2.$$

*Example 2* ( $f = ax^2 + bx + c$ ). This is irreducible of degree 2 exactly when  $a \neq 0$ ,  $\gcd(a, b, c) = 1$ , and  $D := b^2 - 4ac$  is not a square. Since  $(f')^2 = 4af + D$ ,

$$\rho = \text{Res}(f, f') = -aD, \quad u = 4a^2, \quad v = -af',$$

so  $M = -acD$  and

$$F = a^3cD^2x^2 - abDx + 1, \quad W = a^3bcDx^2 - a(b^2 - 2ac)x,$$

$$g = a^6bc^3D^3x^4 - 2a^4c^2(b^2 - ac)D^2x^3 + a^2bc(b^2 - ac)Dx^2 - 2ac(b^2 - 3ac)x,$$

with  $g' = 2ac(2a^2bcDx - (b^2 - 3ac))F$ . For  $b = 0$  the quartic term drops and  $\deg g = 3 = n + 1$ , the least any solution can have. On  $f = x^2 - d$  that cubic is  $g(x) = -D_3(4d^2x, d)/(2d)$ , a rescaling of the Dickson polynomial  $D_3(x, d) = x^3 - 3dx$ , whose pair of critical values is classical.

*Example 3* ( $f = x^3 - x - 1$ ). The construction gives  $M = 23$  and a sextic. Then a search exhibits a quintic. Let  $\theta$  be a root of  $f$ . Then Theorem 1 with  $\beta = 1/(\theta^2 - \theta + 2)$ , so that  $m_\beta = 23x^3 - 23x^2 + 8x - 1$ ,  $P = 46x^2 - 23x + 3$ , and  $Q = -1058x^2 + 667x - 115$ , gives the quintic

$$g = -24334x^5 + 39675x^4 - 26450x^3 + 9085x^2 - 1610x + 118.$$

This is the least possible degree. By Proposition 1 a solution has degree at least 4, and degree 4 cannot happen. In fact no  $g \in \mathbb{Q}[x]$  of degree 4 has  $\theta$  as a critical value. Suppose  $g(\beta) = \theta$  and  $g'(\beta) = 0$  with  $\deg g = 4$ . Up to a rational translation of  $x$ ,  $g'$  is a multiple of a depressed cubic,  $g' = 4a(x^3 + ux + w)$  with  $a, u, w \in \mathbb{Q}$ ,  $a \neq 0$ , so  $g = ah + r$  with  $h = x^4 + 2ux^2 + 4wx$  and  $r \in \mathbb{Q}$ , and the critical values of  $g$  are  $as_i + r$  with  $s_i = h(\beta_i)$ , where the  $\beta_i$  are the roots of  $x^3 + ux + w$ . Since  $\beta_i^3 = -u\beta_i - w$ ,  $s_i = u\beta_i^2 + 3w\beta_i$ , and the characteristic polynomial of this element of  $\mathbb{Q}[\beta]/(\beta^3 + u\beta + w)$  is

$$s(z) = z^3 + 2u^2z^2 + (u^4 + 18uw^2)z + 2u^3w^2 + 27w^4.$$

Now  $s(z)$  is a monic cubic over  $\mathbb{Q}$  with root  $(\theta - r)/a$ , whose minimal polynomial is  $a^{-3}f(az + r)$ , so  $s(z) = a^{-3}f(az + r)$ . The  $z^2$  coefficient gives  $r = 2au^2/3$ , and with it the  $z$  and constant coefficients become  $-1/a^2 = 18uw^2 - u^4/3$  and  $-1/a^3 = 27w^4 - 10u^3w^2 - 2u^6/27$ . The first forces  $u \neq 0$ , so  $\sigma = 27w^2/u^3 \in \mathbb{Q}$ , and the two equations read  $-1/a^2 = -u^4(1 - 2\sigma)/3$  and  $-1/a^3 = u^6(\sigma^2 - 10\sigma - 2)/27$ . Cubing the first and squaring the second gives  $-1/a^6$  and  $1/a^6$  on the left, so adding them,  $u$  and  $a$  drop out:  $(\sigma^2 - 10\sigma - 2)^2 = 27(1 - 2\sigma)^3$ , that is,  $\sigma^4 + 196\sigma^3 - 228\sigma^2 + 202\sigma - 23 = 0$ . This is monic with integer coefficients, so a rational root is an integer dividing 23. But the polynomial takes the values 148 and  $-648$  at  $\sigma = \pm 1$  and is nonzero at  $\pm 23$ . So there is no rational  $\sigma$ , a contradiction.

*Example 4* (Dedekind's cubic).  $f = x^3 - x^2 - 2x - 8$ , the standard example of a field whose ring of integers is not monogenic. Here 2 is a common index divisor [Ded78]. With  $u = 21x - 125$ ,  $v = -7x^2 + 44x - 3$ ,  $\rho = 1006$ ,  $f(0) = -8$ ,  $M = -8048$ ,

$$F = 65158925824x^3 + 8096288x^2 - 2012x + 1, \quad W = -194310912x^3 + 426544x^2 + 358x,$$

$$g = 101288722414414331904x^6 - 209759614012620800x^5 \\ - 217370176548864x^4 - 14767629312x^3 + 2350016x^2 - 10912x.$$

## 4 Computerization

The construction of  $g$  was provided by AI. However, the crucial step that allowed the agent to provide answers was my proof of Corollary 1. Once I recast the problem as the search for an irreducible polynomial  $F$  satisfying the hypotheses of Petrov's theorem, the agent found  $g$  for a general linear polynomial (which one can do by hand), then for a general quadratic, then for a general cubic, and finally gave the formula for  $g$  for arbitrary  $f$ . Previously, all agents I tried failed to construct  $g$  for  $x^3 - x - 1$ . An agent later found Petrov's theorem on MathOverflow after I insisted it should do a deep literature search, including all the classical literature. The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,700 lines [Acu26b]. An online calculator is provided for the reader to verify the construction [Acu26a]. It runs the construction in the browser and, if  $\alpha$  is real, generates a Desmos plot that allows the reader to check visually that it is a critical value. The implemented program does have a virtue that Petrov's construction doesn't have. Because it uses the reduced resultant, the coefficients come out as small as possible.

## References

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